16.07 Dynamics — Quiz 4: Kane Damper (Solutions)

The questions below refer to the rigid body with a Kane damper from Problem Set 4. Recall that the fluid applies a torque $C_d\left({}^B\boldsymbol{\omega}_d - {}^B\boldsymbol{\omega}\right)$ to the body, and that the total kinetic energy changes at the rate

$$\dot{T} = -C_d \left\| {}^B\boldsymbol{\omega}_d - {}^B\boldsymbol{\omega} \right\|^2 .$$

1. Without a damper, spin about which principal axes is stable? Which of these does the Kane damper make unstable, and why?

Answer: Without a damper, both the major and minor axes are stable. The damper makes minor-axis spin unstable. The damper torques are internal, so the total angular momentum $\|\mathbf{h}_{\text{tot}}\|$ is conserved, but energy is dissipated whenever the body nutates ($\boldsymbol{\omega}_d \neq \boldsymbol{\omega}$). For fixed $\|\mathbf{h}_{\text{tot}}\|$, the energy of pure spin about principal axis i is

$$T_i = \frac{\|\mathbf{h}_{\text{tot}}\|^2}{2\,(J_i + J_d)},$$

so minor-axis spin is the *maximum*-energy state and major-axis spin is the *minimum*. Any small nutation about the minor axis loses energy, which carries the body further from minor-axis spin, so the nutation grows until the body reaches the only state the damper can't drain: major-axis spin.

2. What happens to the rate of energy dissipation in the limit $C_d \to 0$? What about the limit $C_d \to \infty$?

Answer: It goes to zero in both limits. As $C_d \to 0$, the damper decouples from the body. The relative rate $\boldsymbol{\omega}_d - \boldsymbol{\omega}$ may be large, but $\dot{T} \to 0$ because of the factor of C_d. As $C_d \to \infty$, the damper locks to the body, so the relative rate shrinks like $1/C_d$ and

$$\dot{T} \sim C_d \cdot \left(\frac{1}{C_d}\right)^2 = \frac{1}{C_d} \to 0.$$

The system then behaves like a single rigid body with inertia $\mathbf{J} + J_d\mathbf{I}$, which conserves energy. So an intermediate value of C_d dissipates energy fastest, and a Kane damper must be tuned.

3. A body with principal inertia matrix $\mathbf{J} = \text{diag}(2,\ 3,\ 4)$ contains a Kane damper with $J_d = 1$. It starts in an equilibrium spin at rate Ω about $\hat{b}_1$ (the minor axis), with the damper co-rotating, and is then given a tiny disturbance. About which axis, and at what angular velocity, does the body eventually spin?

Answer: About the major axis $\hat{b}_3$, with ${}^B\boldsymbol{\omega} = \pm\frac{3}{5}\,\Omega\,\hat{b}_3$. Angular momentum, not energy, is conserved. Initially, the body and the damper both spin at Ω about $\hat{b}_1$, so

$$\|\mathbf{h}_{\text{tot}}\| = (J_1 + J_d)\,\Omega = (2+1)\,\Omega = 3\,\Omega.$$

In the final state, the body and the damper both spin at ω_3 about $\hat{b}_3$, so

$$(J_3 + J_d)\,|\omega_3| = 3\,\Omega \implies |\omega_3| = \frac{3\,\Omega}{4+1} = \frac{3}{5}\,\Omega.$$

The sign depends on the details of the disturbance, so either sign is correct. Physically, the body turns over so that $\hat{b}_3$ lines up with the inertially fixed angular momentum vector, along the original direction of $\hat{b}_1$. The kinetic energy drops by a factor of $(J_1 + J_d)/(J_3 + J_d) = 3/5$.